CA Note

1 Basic Knowledge

Group $(G, *)$**: G1**: Closure **G2**: Associativity **G3**: Identity element **G4**: Inverse element

Ring $(R, +, \cdot)$**: R1**: $(R, +)$ is an abelian group **R2**: $(R, \cdot)$ is monoid. (no inverse) **R3**: Distributive law (left and right)

Ideal (I)**: I1**: $\neq \emptyset$ **I2**: $\forall a, b \in I, a - b \in I$ **I3**: $\forall r \in R, i \in I, ri, ir \in I$ **Lemma:** $I = R \Leftrightarrow 1 \in I \Leftrightarrow I \cap R^{\times} \neq \emptyset$

Subgroup (H)**: H1**: $e \in H$ **H2**: $\forall a, b \in H, ab \in H$ **H3**: $\forall a \in H, a^{-1} \in H$ **Subring** (S)**: S1**: $1 \in S$ **S2**: $\forall a, b \in S, a - b \in S$ **S3**: $\forall a, b \in S, ab \in S$

Integral Domain (ID): ID1: communicative ring **ID2**: no zero divisors **Lemma:** Finite ID $\Rightarrow$ field ID $\Rightarrow$ Cancellation Law

Factor Ring (R/I): $a + I = b + I \Leftrightarrow a - b \in I$ $(R/I, \oplus, \odot)$ is a ring. ($1_{R/I} = 1_R + I, 0_{R/I} = 0_R + I, I$ ideal)

Ring Homomorphism: $\varphi : R \to S$ s.t. **RH1**: $\varphi(a + b) = \varphi(a) + \varphi(b)$ **RH2**: $\varphi(ab) = \varphi(a)\varphi(b)$ **RH3**: $\varphi(1_R) = 1_S$

· **1**: Ker(φ) is an ideal. **2**: Im(φ) is a subring. **3**: φ is 1-1 $\Leftrightarrow$ Ker$(\varphi) = \{0_R\}$. **FIT:** $R/$ Ker$(\varphi) \cong$ Im(φ) (First Isomorphism Theorem)

· **Canonical Homomorphism:** $can : R \to R/I$ s.t. $can(r) = r + I$ is a onto ring Homomorphism. Ker$(can) = I$ $\updownarrow$ ($\exists! \tilde{\varphi}$ 双射)

· **Lemma:** If $\varphi : R \to S$ ring Homomorphism s.t. $I \subseteq$ Ker(φ), then $\exists! \tilde{\varphi} : R/I \to S$ s.t. $\varphi = \tilde{\varphi} \circ can$ (i.e. $\tilde{\varphi}(r + I) = \varphi(r)$)

Lagrange's Theorem: G finite group, $H \leq G$ subgroup, then $|H| \mid |G|$

2 Types of Ideals

Division: If R ring, then: $\forall f, g \in R[x],_{g \text{ 首相系数可逆}} \Rightarrow \exists q, r \in R[x]$ s.t. $f(x) = g(x)q(x) + r(x)$ and deg$(r) <$ deg(g) or $r = 0$.

Lemma: If R ring, $g \in R[x], g \neq 0, I = (g)$. $\Rightarrow$ [1] $\forall i \in R[x]/I, i = f + I$, deg$(f) <$ deg(g)

g 首相系数可逆 [2] If $a, b \in R[x]$, deg$(a, b) <$ deg(g) and $a + I = b + I$, then $a = b$. (系数也相同)

Remark: 有些时候我们能用 gcd, lcm 计算 $< a, b > = <$ gcd$(a, b) >$ 和 $a \cap b = <$ lcm$(a, b) >$ [If R communicative ring] 但注意! 要在能够用 Division Algorithm | 有 gcd, lcm 的环里用.

Irreducible: If R in ID, $0 \neq a \in R$ (not unit) is irreducible $\Leftrightarrow$ $a \neq bc$ for any non-units $b, c \in R$

Principal Ideal: Ideal I: Iff $I = aR$ **Prime Ideal:** Ideal I**≠R**: $ab \in I \Rightarrow a \in I$ or $b \in I$. (OR: If $a \notin I$ and $b \notin I$, then $ab \notin I$)

Maximal Ideal: Ideal I**≠R**: If J ideal and $I \subseteq J \subseteq R$, then $J = I$ or $J = R$.

Radical|Radical Ideal: I: **Radical**: $\sqrt{I} = \{r \in R : r^n \in I$ for some $n \in \mathbb{N}\}$ **Radical Ideal**: If $I = \sqrt{I}$ $I \subseteq \sqrt{I}$ always.

Properties: Relationship between these kinds of ideals: **Maximal Ideals $\subseteq$ Prime Ideals $\subseteq$ Radical Ideals $\subseteq$ Ideals**

1. R communicative ring, I ideal. **I prime $\Leftrightarrow R/I$ is an ID.** **I maximal $\Leftrightarrow R/I$ is a field** **I maximal $\Rightarrow I$ prime**
2. R ring, I ideal, J ideal of R/I, $T = \{r \in R : r + I \in J\}$(reps of J). Then [1] T**: Ideal of** R [2] $I \subseteq T$ [3] $J = \{t + I : t \in T\}$ [4] $(R/I)/J \cong R/T$
3. R communicative ring, I ideal. **I radical $\Leftrightarrow R/I$ has no nilpotent elements** except 0 **I prime $\Rightarrow I$ radical** (nilpotent: $r^n = 0$)
4. If R (commutative with *unity*), then any ideal $I \neq R$ is contained in a *maximal* ideal. If r is *not a unit*, then $1 \notin \langle r \rangle = rR$

3 Polynomial Rings

Polynomial Ring: R *communicative* ring. $R[x_1, ..., x_n]$ is a ring. The elements of it: $\sum_{\alpha} a_{\alpha} x^{\alpha}$ $\alpha = (\alpha_1, ..., \alpha_n), a \in R, x^{\alpha} = x_1^{\alpha_1} ... x_n^{\alpha_n}$

- **Degree**: deg$(f) = \max\{\sum_i \alpha_i : a_{\alpha} \neq 0\}$ If $f = 0$, deg$(f) = -\infty$. **2nd Definition**: $R[x, y] = R[x][y]$ and so on. (induction 喜欢)
- deg$(fg) \leq$ deg$(f) +$ deg(g) unless R is an ID. If R is an ID, then $R[x_1, ..., x_n]$ is an ID.